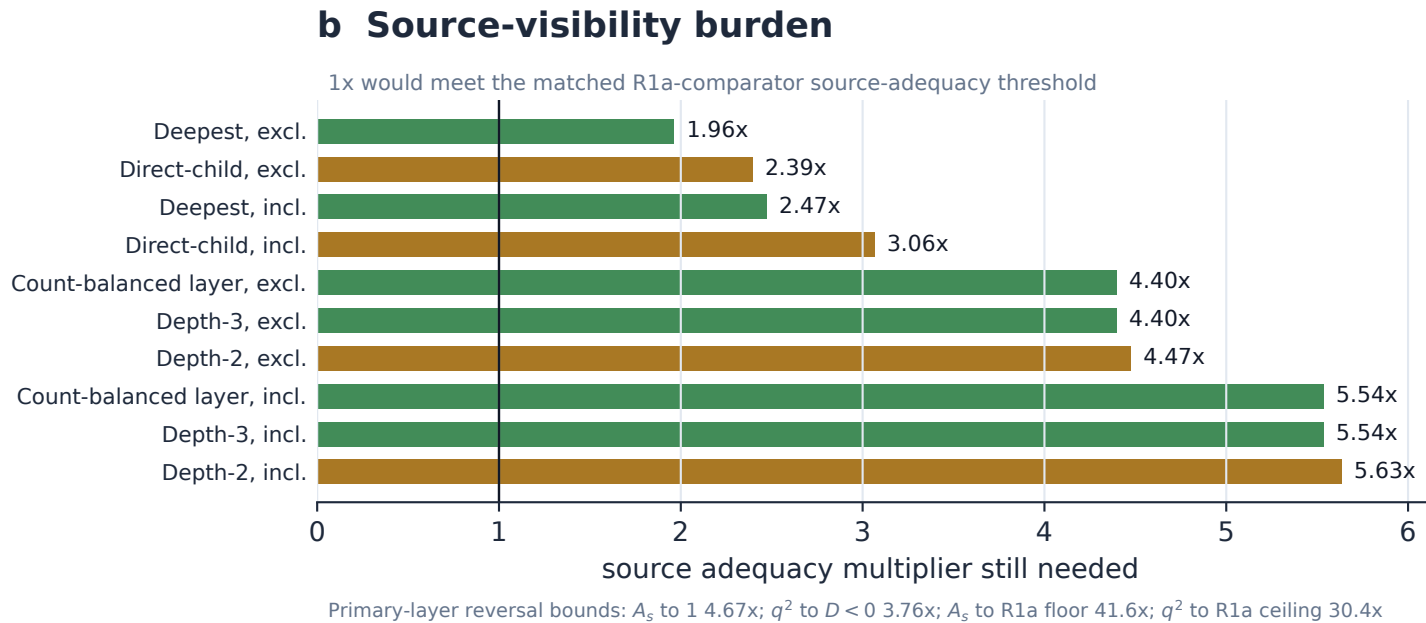
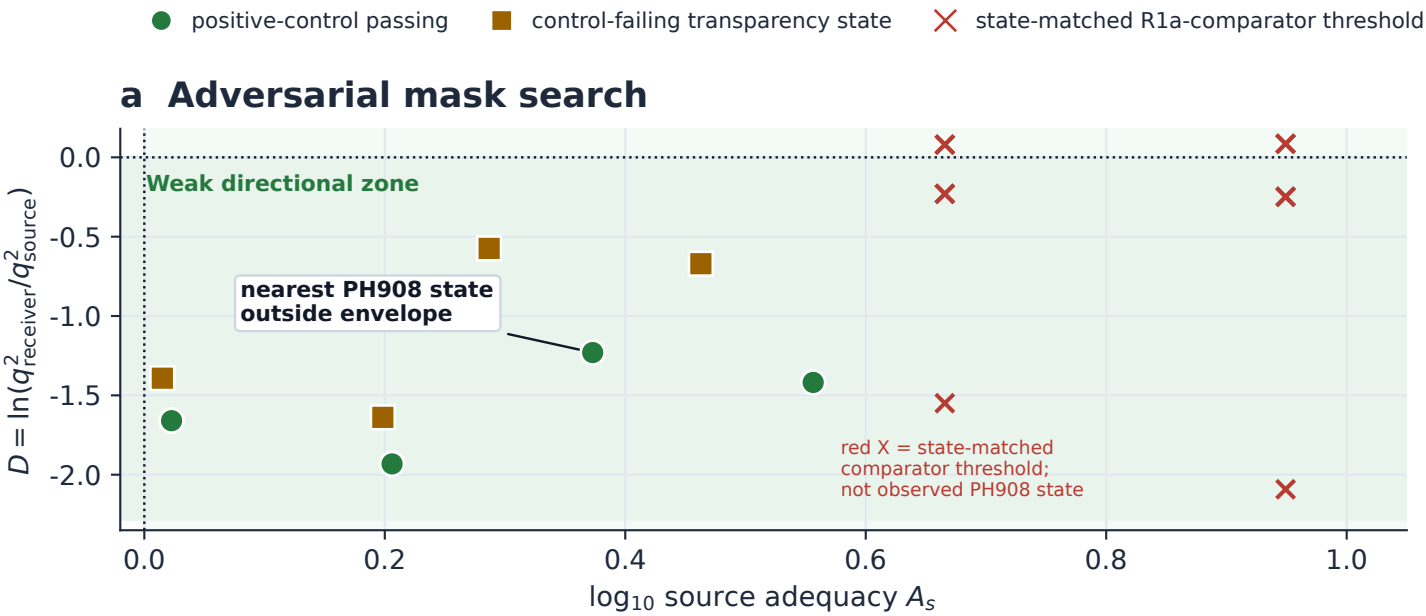
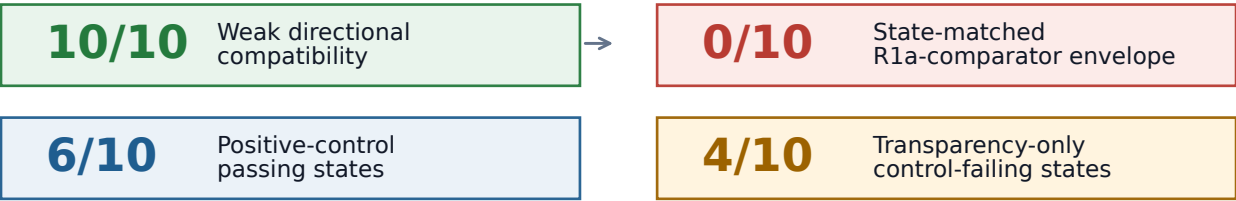


Visibility and mask-choice stress tests do not place PH908 in the matched R1a-comparator envelope

Visibility-bias reversal, synthetic degradation controls, and adversarial masks are shown as model checks, not origin evidence.

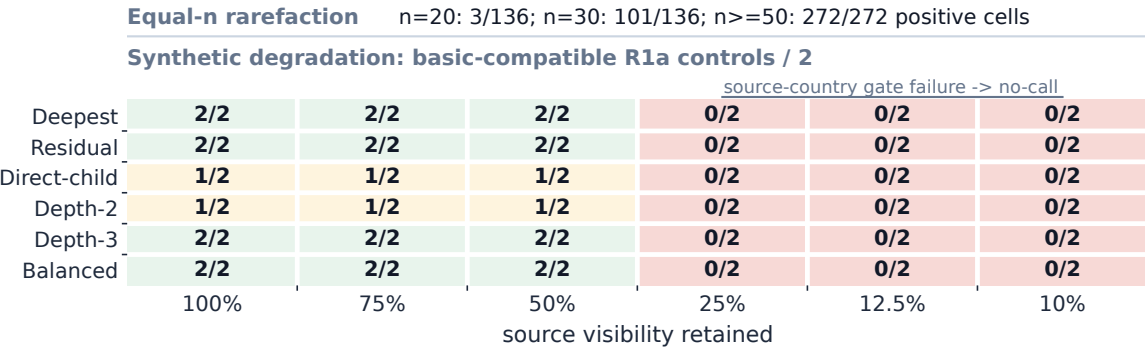


c Adversarial decision gate



Interpretation: mask flexibility can create weak directional compatibility, but the strict matched-comparator gate remains closed.

d Rarefaction and synthetic degradation controls



Sources: Source data are provided with the numbered Figure 4 source-data bundle. Weak directional compatibility is $A_s > 1$ and $D < 0$; matched-envelope entry additionally requires the state-matched R1a-comparator threshold. At $\leq 25\%$ retained source visibility, compatible comparators fail the source-country visibility gate in the deterministic degradation control. Equal-n rarefaction summarizes power and denominator limits; visibility burden and synthetic degradation controls are operating-characteristic checks, not population-size estimates. The figure does not identify PH908 geographic origin, ethnic identity, population continuity, migration route, or any population-level probability.